

Lab Title: Digital Currency and Chrome Web History: Cryptocurrency Transaction Reconstruction

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Course Name: Computer Forensics Case Study II (CIP-B105)

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Instruction Name: Aminu Idris

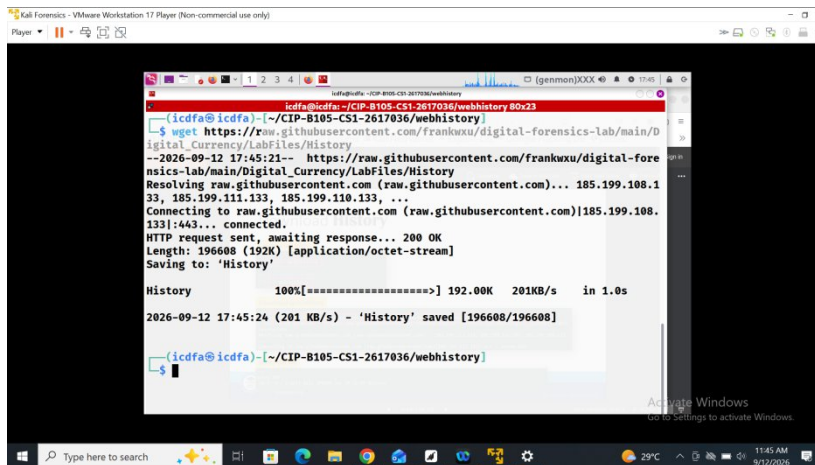
Platform Used: ICDFA Kali Linux and Wireshark

Executive Summary

This investigation looked at Chrome browser activity linked to an alleged online drug transaction. The evidence was preserved, hashed, and reviewed from the SQLite databases, with timestamps converted to UTC. The activity was then pieced together across Craigslist, Gmail, Imgur, downloads, and Bitcoin-related sites.

Overall, the evidence supports several parts of the seven-stage hypothesis, although browser history alone cannot prove that every action actually occurred. The timeline and evidence matrix show what is supported by the evidence and where the limitations are.

1. I created the lab structure and downloaded the Logs to be analyzed

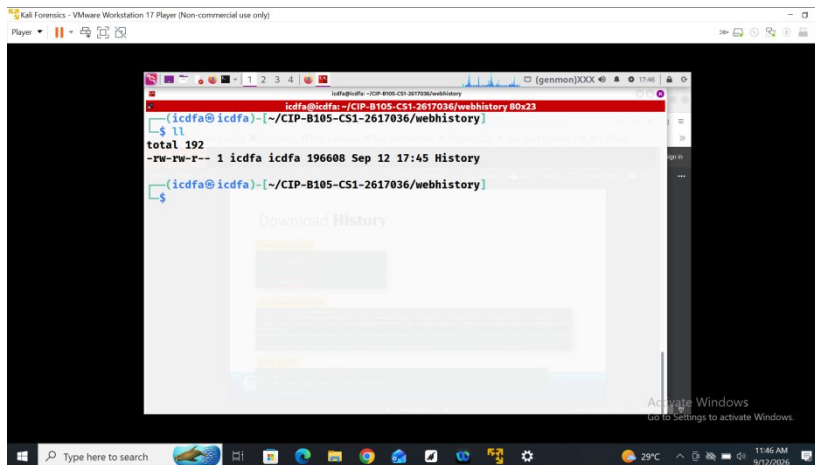


```
icdfa@icdfa:~/CIP-B105-CS1-2617036/webhistory 80x23
$ wget https://raw.githubusercontent.com/frankxuxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History
--2026-09-12 17:45:21-- https://raw.githubusercontent.com/frankxuxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 196608 (192K) [application/octet-stream]
Saving to: 'History'

History      100%[=====] 192.00K  201KB/s   in 1.0s
2026-09-12 17:45:24 (201 KB/s) - 'History' saved [196608/196608]

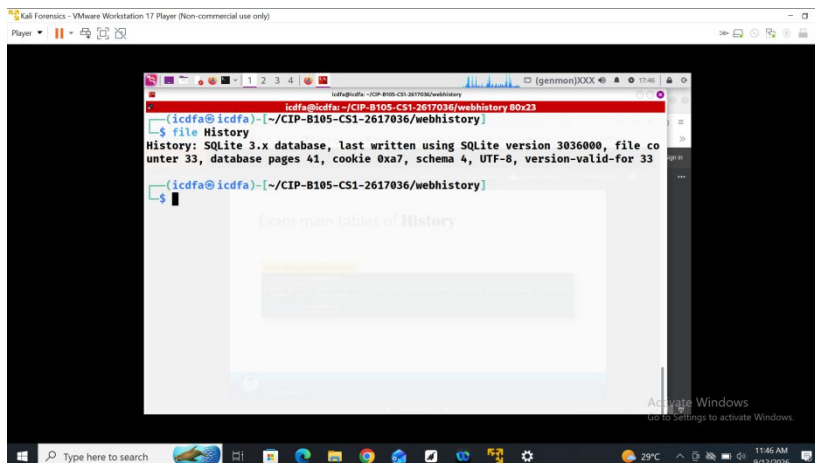
icdfa@icdfa:~/CIP-B105-CS1-2617036/webhistory
$
```

2. Then I verified it



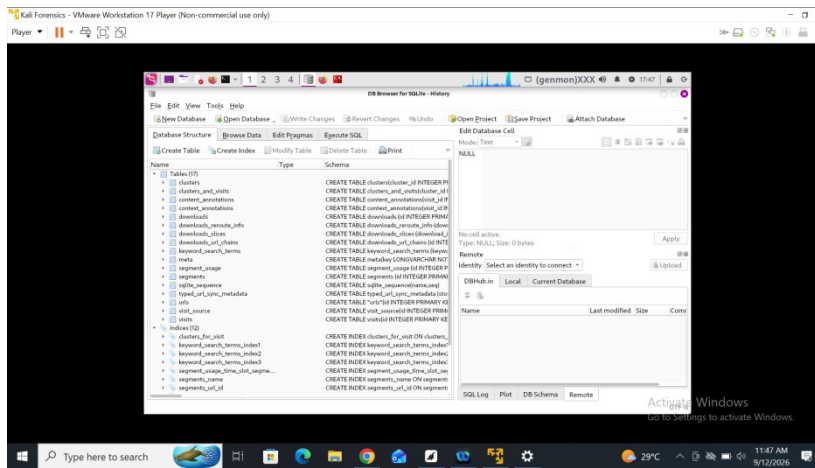
```
icdfa@icdfa:~/CIP-B105-CS1-2617036/webhistory 80x23
$ ll
total 192
-rw-r--r-- 1 icdfa icdfa 196608 Sep 12 17:45 History
icdfa@icdfa:~/CIP-B105-CS1-2617036/webhistory
$
```

3. File History result

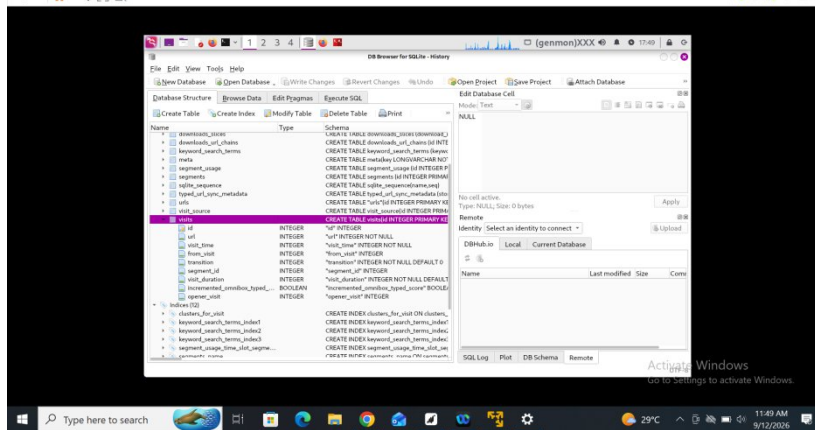
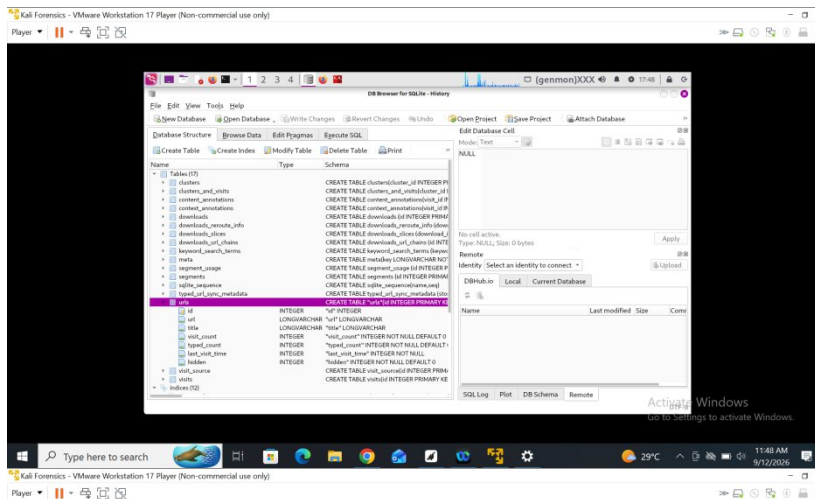


```
icdfa@icdfa:~/CIP-B105-CS1-2617036/webhistory 80x23
$ file History
History: SQLite 3.x database, last written using SQLite version 3036000, file co
unter 33, database pages 41, cookie 0xa7, schema 4, UTF-8, version-valid-for 33
icdfa@icdfa:~/CIP-B105-CS1-2617036/webhistory
$
```

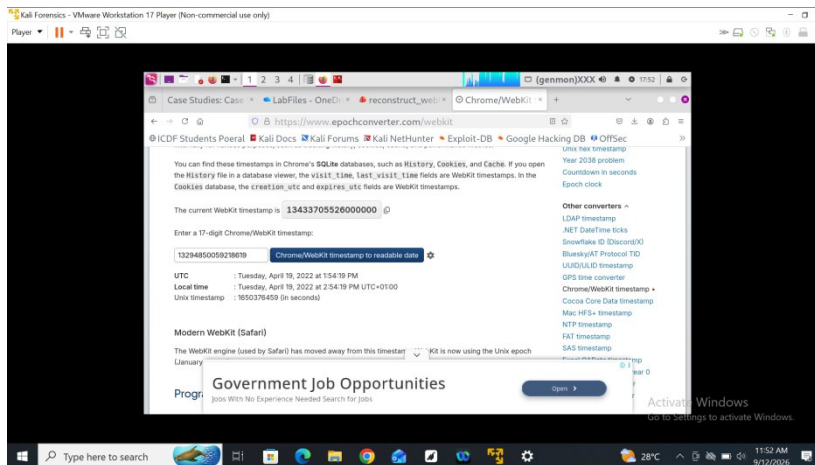
4. Then I opened the log with DB browser



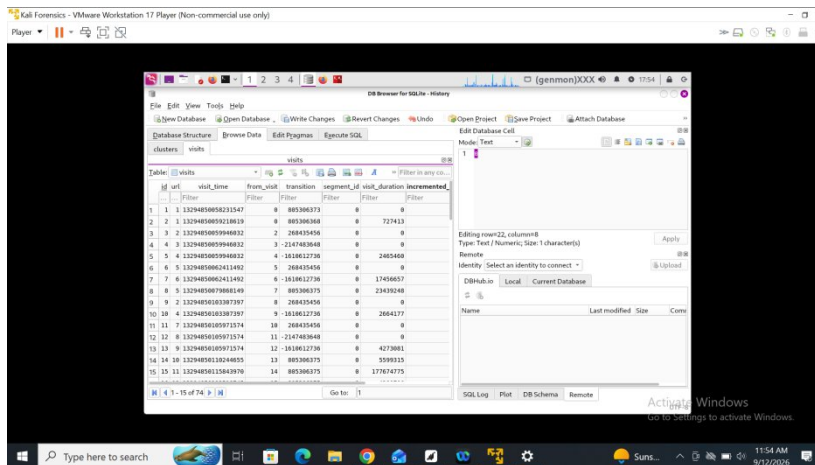
5. The urls, visits and the download schema



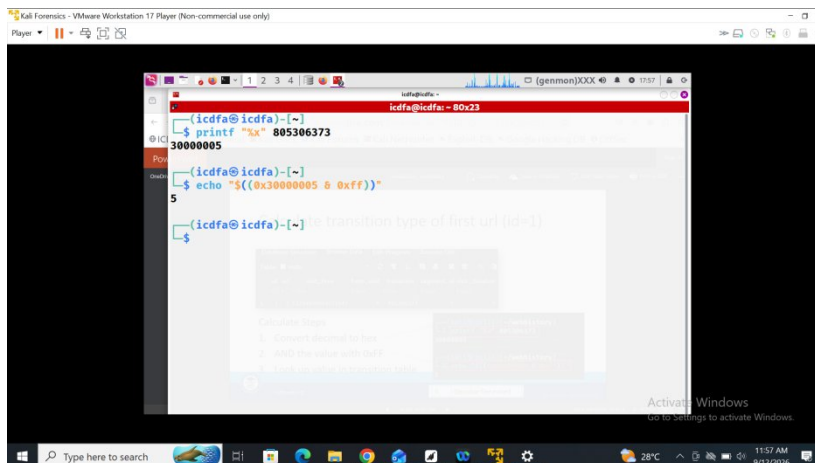
6. Converting timestamp with epochconverter/webkit to show realtime information



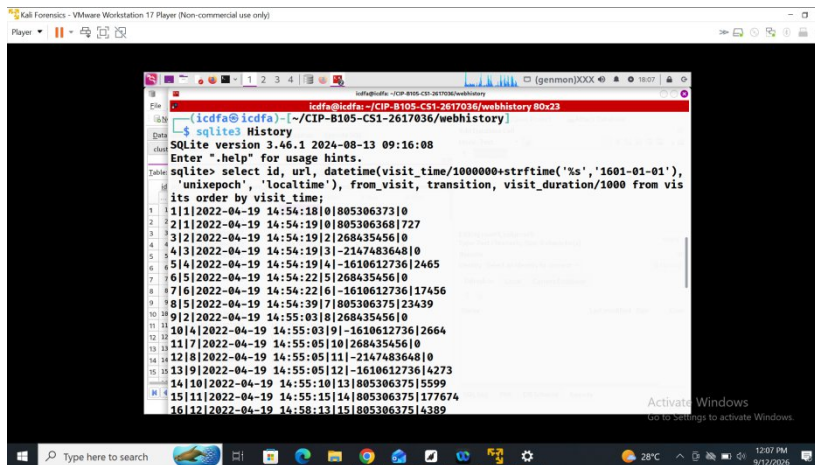
7. The information in the visit section



8. Calculating transition type

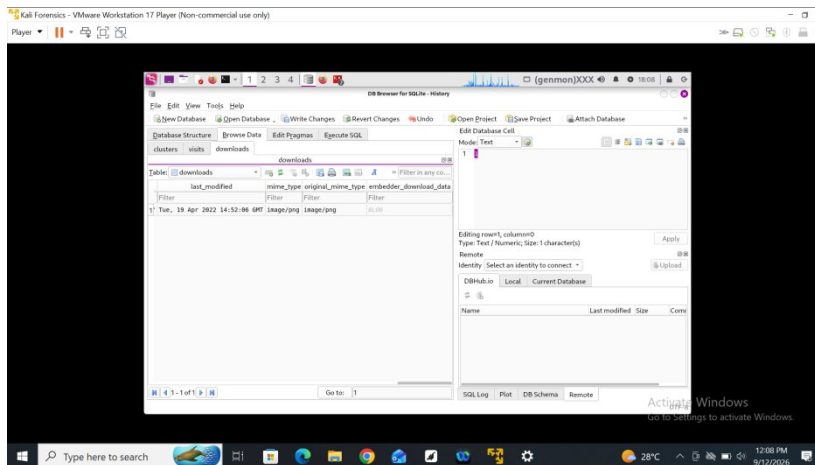


9. Sqlite3 History to get significant information



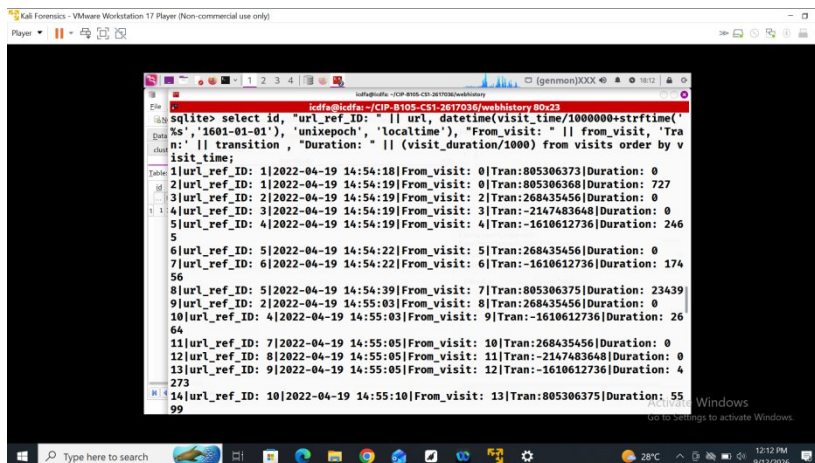
```
icdfa@icdfa:~/CIP-8105-CS1-2617036/webhistory$ sqlite3 History
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> select id, url, datetime(visit_time/1000000+strftime('%s','1601-01-01'),
'unixepoch', 'localtime'), from_visit, transition, visit_duration/1000 from vis
its order by visit_time;
1|1|2022-04-19 14:54:18|0|805306373|0
2|1|2022-04-19 14:54:19|0|805306368|727
3|2|2022-04-19 14:54:19|2|268435456|0
4|3|2022-04-19 14:54:19|3|-2147483648|0
5|4|2022-04-19 14:54:19|4|-1610612736|2465
6|5|2022-04-19 14:54:22|5|268435456|0
7|6|2022-04-19 14:54:22|6|-1610612736|17456
8|5|2022-04-19 14:54:39|7|805306375|23439
9|2|2022-04-19 14:55:03|8|268435456|0
10|4|2022-04-19 14:55:03|9|-1610612736|2664
11|7|2022-04-19 14:55:05|10|268435456|0
12|8|2022-04-19 14:55:05|11|-2147483648|0
13|9|2022-04-19 14:55:05|12|-1610612736|4273
14|10|2022-04-19 14:55:10|13|805306375|5599
15|11|2022-04-19 14:55:15|14|805306375|177674
16|12|2022-04-19 14:58:13|15|805306375|14389
```

10. The download information



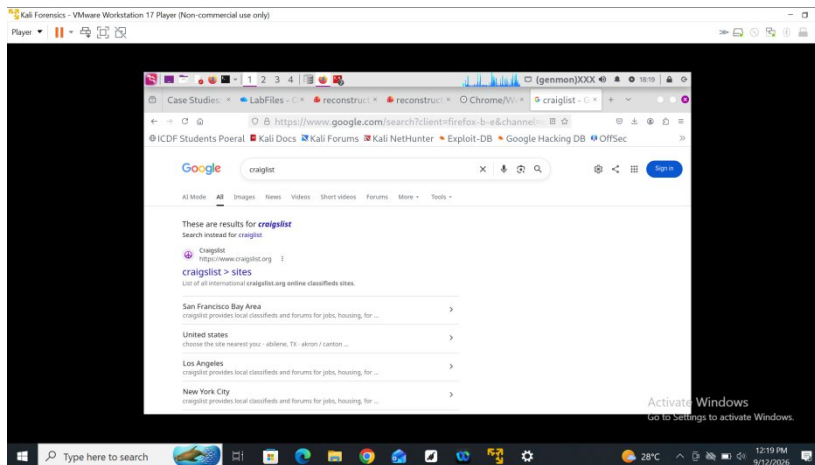
File Name	Size	Date
last_modified		
name_type original_name_type embedder_download_data		
Filter	Filter	Filter
Tue, 19 Apr 2022 14:52:08 GMT image/png image/png	15.0K	

11. The url ref ID

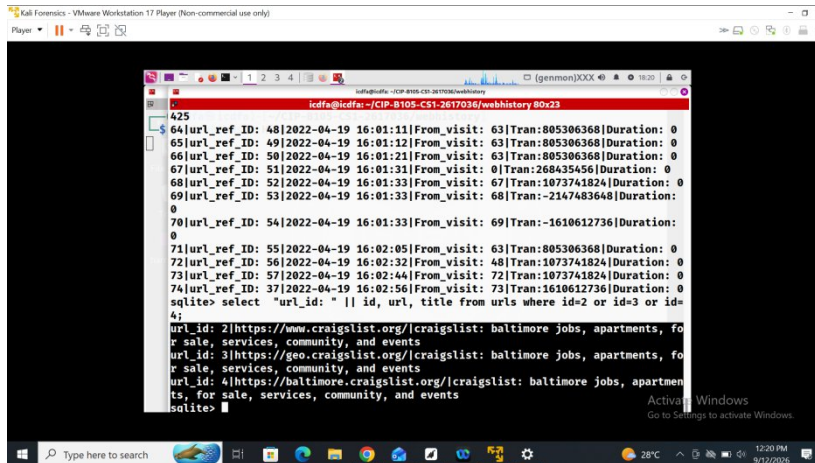


```
icdfa@icdfa:~/CIP-8105-CS1-2617036/webhistory$ sqlite3 History
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> select id, url_ref_ID, url, datetime(visit_time/1000000+strftime('%s','1601-01-01'),
'unixepoch', 'localtime'), "From_visit: " || from_visit, "Tra
n: " || transition, "Duration: " || (visit_duration/1000) from visits order by v
isit_time;
1|url_ref_ID: 1|2022-04-19 14:54:18|From_visit: 0|Tran:805306373|Duration: 0
2|url_ref_ID: 1|2022-04-19 14:54:19|From_visit: 0|Tran:805306368|Duration: 727
3|url_ref_ID: 2|2022-04-19 14:54:19|From_visit: 2|Tran:268435456|Duration: 0
4|url_ref_ID: 3|2022-04-19 14:54:19|From_visit: 3|Tran:-2147483648|Duration: 0
5|url_ref_ID: 4|2022-04-19 14:54:19|From_visit: 4|Tran:-1610612736|Duration: 246
5
6|url_ref_ID: 5|2022-04-19 14:54:22|From_visit: 5|Tran:268435456|Duration: 0
7|url_ref_ID: 6|2022-04-19 14:54:22|From_visit: 6|Tran:-1610612736|Duration: 174
56
8|url_ref_ID: 5|2022-04-19 14:54:39|From_visit: 7|Tran:805306375|Duration: 23439
9|url_ref_ID: 2|2022-04-19 14:55:03|From_visit: 8|Tran:268435456|Duration: 0
10|url_ref_ID: 4|2022-04-19 14:55:03|From_visit: 9|Tran:-1610612736|Duration: 26
64
11|url_ref_ID: 7|2022-04-19 14:55:05|From_visit: 10|Tran:268435456|Duration: 0
12|url_ref_ID: 8|2022-04-19 14:55:05|From_visit: 11|-2147483648|Duration: 0
13|url_ref_ID: 9|2022-04-19 14:55:05|From_visit: 12|Tran:-1610612736|Duration: 4
273
14|url_ref_ID: 10|2022-04-19 14:55:10|From_visit: 13|Tran:805306375|Duration: 55
99
```

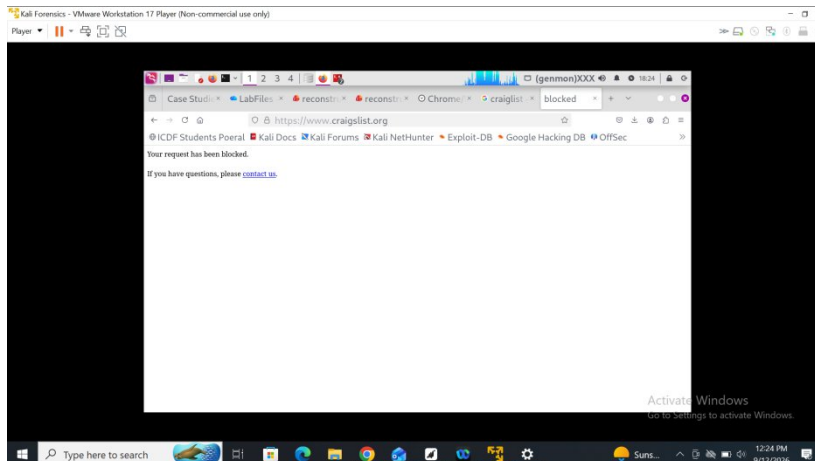

12. Craigslist on google



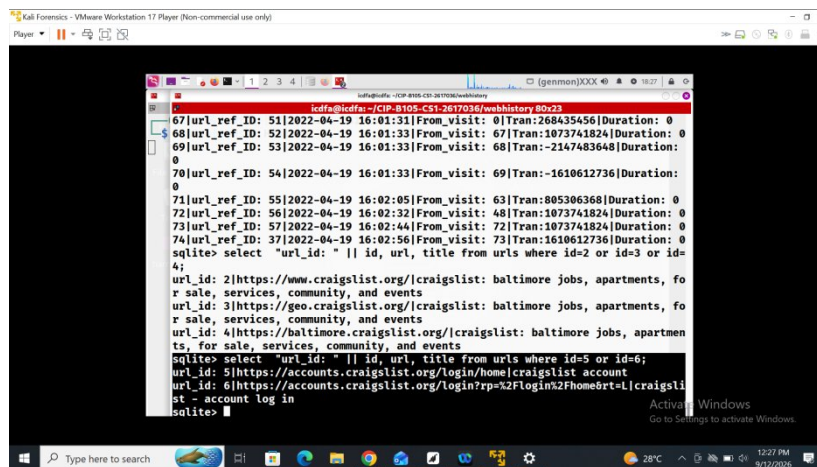
13. Url id 3 to 4



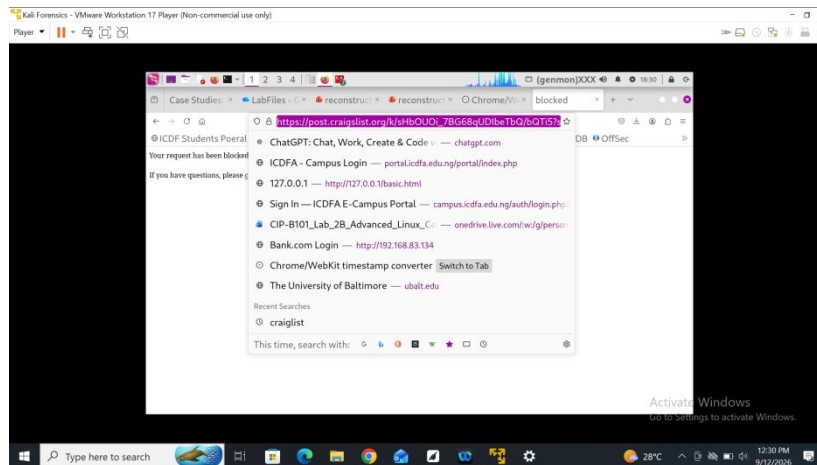
14. Craigslist site was blocked and I was unable to access it



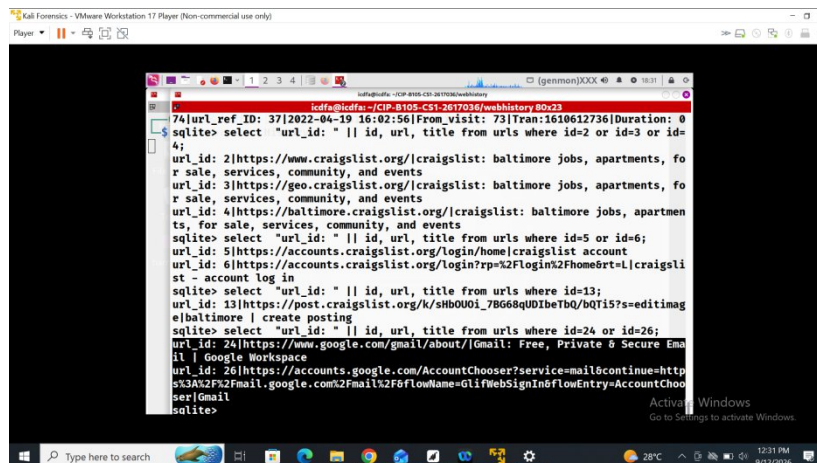
15. Url id 5 to 6



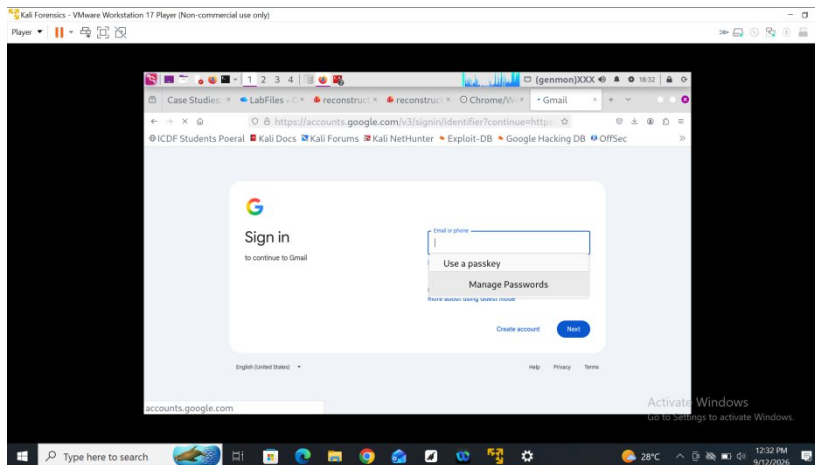
16. I tried the other link and it was still blocked



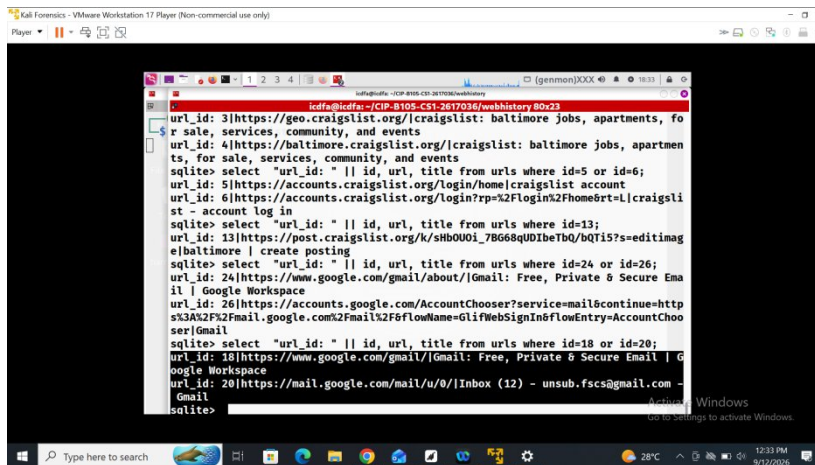
17. Url id 24 to 26



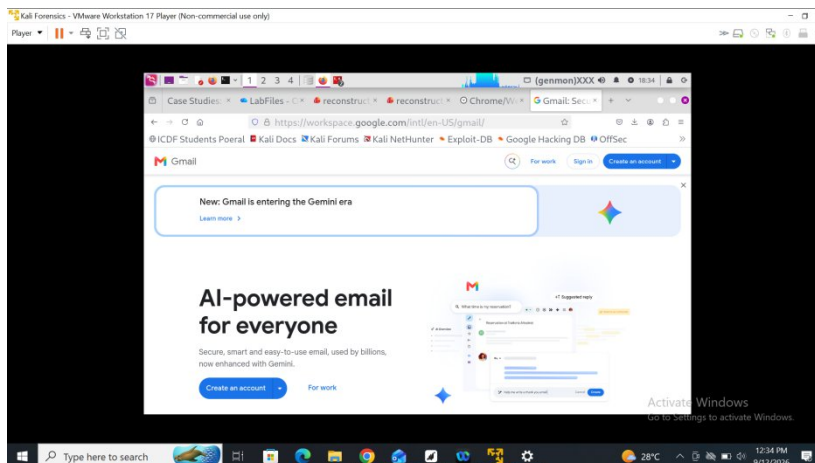
18. The google link



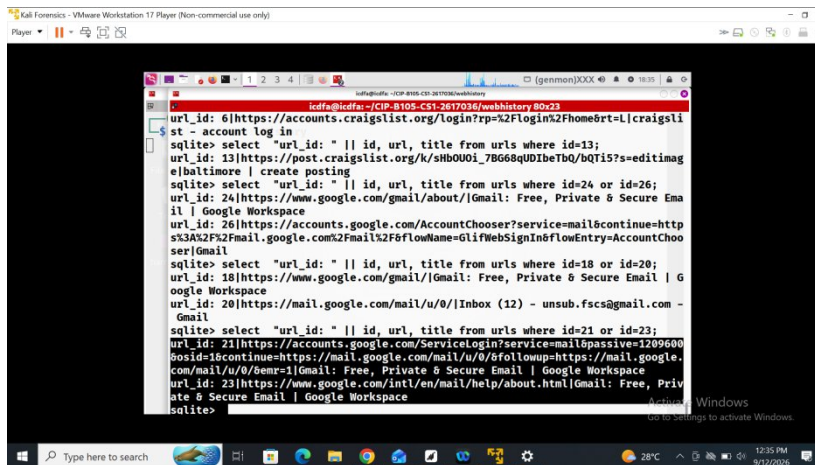
19. Url id 18 to 20



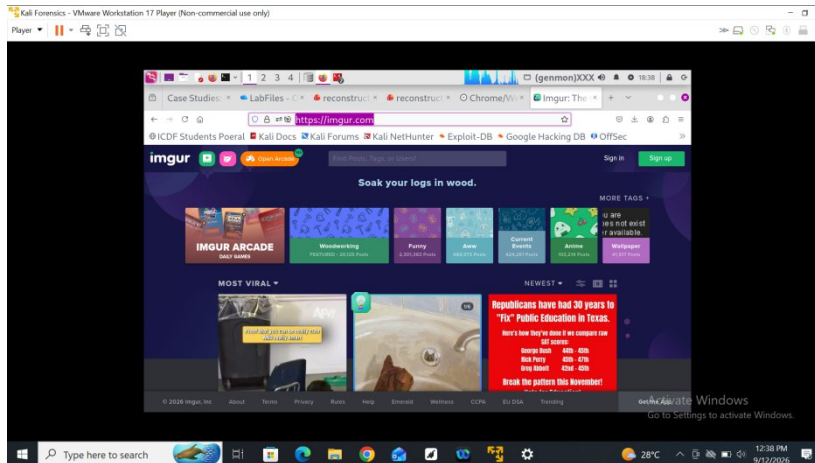
20. Google workspace in the browser



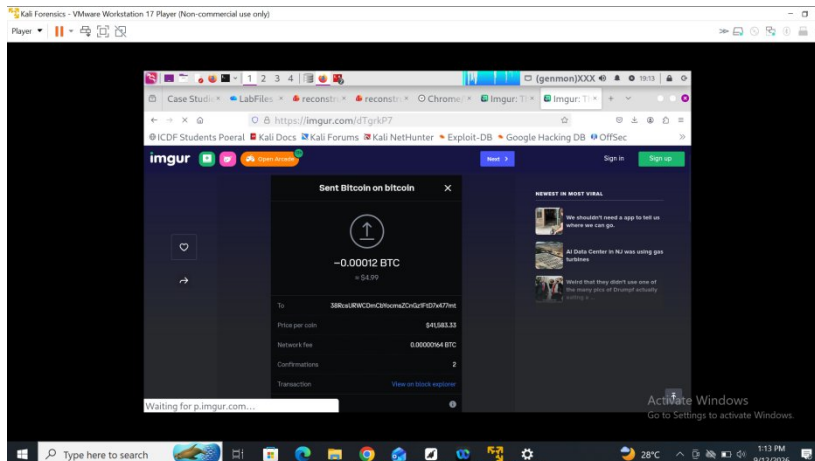
21. Url id 21 to 23



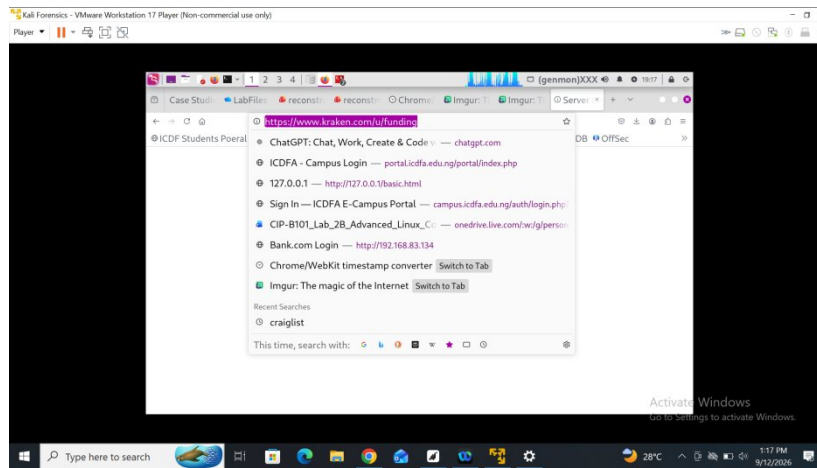
22. Imgur link from the download information table



23. The proof of bitcoin payment



24. Kraken was not loading as well



Conclusion

The investigation successfully reconstructed much of the browser activity and found evidence supporting several stages of the suspected transaction. However, some parts of the investigation could not be fully completed because certain websites were blocked and some URLs were no longer accessible. As a result, some findings could not be independently verified. Despite these limitations, the available browser history, downloads, and related records provided useful evidence for assessing the seven-stage hypothesis.